

Statement of Work (SOW)

Project Title: Spacecraft Structures (SCS): From Concept to Launch Training

Revision: 1.0

Date: April 29, 2026

1.0 Introduction

The NASA/MSFC requires a comprehensive technical short course focused on the design, analysis, and verification of spacecraft structures. The goal is to strengthen the engineering workforce's understanding of structural requirements, failure modes, and risk assessment to reduce total mission cost and ensure mission success.

2.0 Scope of Work

The Contractor shall provide an expert instructor to deliver the **Spacecraft Structures (SCS)** course. The course shall cover the full life cycle of spacecraft structural development, from initial concept through launch.

Key Technical Domains to be Covered:

- Launch environments (Quasi-static, Transient, Random Vibration, and Shock).
- Structural integrity and stress analysis (Factors of Safety, Margins of Safety).
- Finite Element Analysis (FEA) applications and quality assurance.
- Preliminary design and configuration development.
- The Loads-Cycle process and Coupled Loads Analysis (CLA).
- Verification, quality assurance, and final risk assessment.

3.0 Requirements and Deliverables

The Contractor shall fulfill the following requirements:

- **3.1 Course Delivery:** Deliver a 3-day (24-hour) intensive training program.
- **3.2 Instructor Qualifications:** The instructor must be a subject matter expert with a minimum of 30 years of experience in aerospace structural engineering and a proven track record of consulting for NASA or similar space agencies.

- **3.3 Training Materials:** Provide each participant with a digital or printed copy of the [SCS Course Book](#) (latest revision: September 2025).
- **3.4 Curriculum Content:** The curriculum must remain consistent with established industry principles, including the "Ten Principles for Effective Engineering in the Space Industry" as outlined in the course syllabus.

4.0 Technical Criteria for Success (Verification)

The course shall be deemed successful upon completion of the following:

1. **Attendance:** Delivery of the course to the agreed-upon number of NASA engineers and managers.
2. **Knowledge Transfer:** Completion of all eight primary modules, including "Improving the Loads-Cycle Process" and "Final Verification and Risk Assessment."
3. **Feedback:** Collection of participant evaluations to ensure the "Overall Objective" of helping attendees become better engineers is met.

5.0 Period of Performance

The training shall be conducted over three consecutive business days, with the specific dates to be determined by the Contracting Officer's Representative (COR) and the Contractor.

6.0 Location

The course will be held [In-person at Marshall Space Flight Center], as specified in the individual task order.

7.0 References

- Sarafin, Thomas P., ed. *Spacecraft Structures and Mechanisms: From Concept to Launch*.
- NASA-STD-5001, *Structural Design and Test Factors of Safety for Spaceflight Hardware*.
- NASA-STD-5020, *Requirements for Threaded Fastening Systems in Spaceflight Hardware*